Separating Intervention Search from Response Prediction in Closed-Loop Biological Agents

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Abstract

A closed-loop biological agent chooses interventions, observes their responses, and uses those observations to choose subsequent experiments and a final intervention. Does finding a successful intervention also show that feedback improved its predictions of interventions it has not tried? We measure these outcomes separately under an eight-assay budget by scoring held-out intervention forecasts before and after feedback and scoring the final intervention independently. Across 232 tasks from 39 published Boolean networks, replacing an intervention-independent mean predictor with ridge regression lowers held-out squared prediction error by .056 (95% interval $[-.064, -.049]$) while assays and final choices remain fixed. Conversely, changing only the final-choice rule lowers intervention success by 5.0 percentage points $[-9.4, -.7]$ while predictions remain fixed. The intervention-independent mean nevertheless achieves the lowest prediction error among six procedures while distinguishing no interventions, so we additionally measure whether forecasts remain aligned with the correct interventions after their identities are shuffled. In an LLM study, maintaining a hypothesis note increases this alignment on named simulation tasks, but the difference shrinks when gene names are hidden or runs requiring action fallbacks are excluded. In retrospective replay of six human T-cell cytokine screens and K562 combined CRISPRa perturbations, hypothesis notes do not consistently reduce final prediction error relative to observation notes given identical assay histories. These results show that intervention success and response prediction answer different questions about a closed-loop biological agent and should be reported separately; they do not establish a general benefit from hypothesis notes or prospective biological discovery.

1 Introduction

A closed-loop biological agent selects a perturbation, observes its response, and uses that feedback to choose the next experiment. A successful final choice alone cannot establish that predictions of untried perturbations improved. We measure both outcomes within the same experimental budget and identify which part of the decision procedure changes each one.

The procedure receives the allowed interventions and disclosed task information; other outcomes remain hidden until assayed. An *assay* reveals one intervention's response; a *readout* is an output such as cytokine-screen enrichment or simulated gene activity. An *episode* contains eight assays and, where a search objective is defined, a final choice. The procedure commits forecasts before feedback and predicts reserved interventions, called *probes*, before and after the episode. Probe outcomes remain hidden. We score prediction error, alignment with the correct interventions, and final-intervention success separately.

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The distinction is visible in a three-node system. Let u and v be intervention inputs, let y be the target readout, and start all nodes at zero. After one step, $y_1 = u_0 \wedge \neg v_0$. Two procedures choose $(u, v) = (1, 0)$ and both reach $y_1 = 1$. One predicts $u \wedge \neg v$; the other predicts u. Only the first predicts the untried intervention $(1, 1)$ correctly (Figure 1).

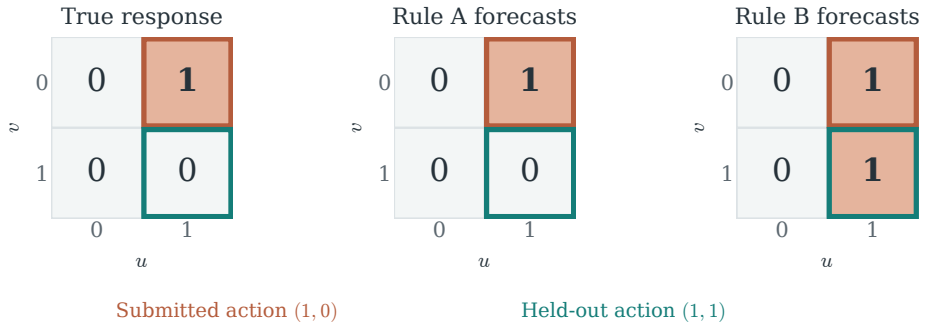


Figure 1: The same successful intervention can accompany different predictions. Both procedures choose $(u, v) = (1, 0)$ and reach the target (rust outline). On the reserved intervention $(1, 1)$ (teal outline), only prediction rule A gives the true response. Cells give y_1 for each input pair.

Boolean-network tests replace the predictor or final-choice rule while fixing the other components. The unchanged outcomes follow from those fixed components; measured differences quantify the replacements' effects. Separate LLM studies compare hypothesis notes with no carried note in simulation and with observation notes on identical histories in measured-data replay. Replay tests the protocol on historical lab measurements, not prospective discovery.

Relation to prior work. BioDiscoveryAgent searches for perturbations with desired phenotypes; the Virtual Lab and Robin connect LLMs to experimental design and feedback [Roohani et al., 2025, Swanson et al., 2025, Ghareeb et al., 2026]. GEARS and other response predictors estimate outcomes of genetic perturbations [Lotfollahi et al., 2023, Roohani et al., 2024, Cui et al., 2024]. Simple linear predictors can rival larger models, and Systema shows why common response components can inflate prediction scores [Ahlmann-Eltze et al., 2025, Viñas Torné et al., 2025]. Feedback-permutation controls are also established [Gupta et al., 2025]. Our contribution is an evaluation protocol for closed-loop biological agents that isolates prediction changes and search outcomes under a shared assay budget. We vary assay selection, response prediction, and final-choice rules, and report held-out error, intervention alignment, and final-intervention success separately. The predictors and feedback controls are established tools. Appendix G gives the focused novelty comparison.

2 Problem formulation

In the simulation experiments, each task fixes a Boolean network with n named nodes, each either active (1) or inactive (0). The evaluator knows its transition function $F : \{0,1\}^n \to \{0,1\}^n$; the procedure does not. The procedure sees the initial state $x^{(0)}$, horizon $H \geq 1$, a readout set R, and a signed interaction graph when one is available. The graph shows possible influences but does not specify F. An intervention a clamps at most two eligible nodes: their values are held at 0 or 1 throughout the assay. Every assay starts from $x^{(0)}$ with the clamp imposed and runs H synchronous steps:

$$x_{a,i}^{(t+1)} = \begin{cases} a_i, & i \in \mathrm{dom}(a), \\ F_i(x_a^{(t)}), & i \notin \mathrm{dom}(a), \end{cases} \qquad t = 0, \ldots, H-1, \tag{1}$$

The observation is $o(a) = x_{a,R}^{(H)} \in \{0,1\}^{|R|}$. Assays reset the physical state, so only the procedure's information changes between rounds.

The task supplies target bits g on $G \subseteq R$. Let $\mathcal{A}$ be the legal actions, which cannot clamp target nodes. After eight distinct assays, the procedure submits $a_{\mathrm{final}} \in \mathcal{A}$ and is scored for reaching the target

$$S_{\mathrm{goal}} = \mathbf{1}\big[o_G(a_{\mathrm{final}}) = g\big], \qquad a_{\mathrm{final}} \in \mathcal{A}. \tag{2}$$

At round t, the procedure uses the disclosed task and history $h_t = ((a_s, p_s, o(a_s)))_{s<t}$ to choose an untried assay $a_t \in \mathcal{A}$ and forecast each readout with probability $p_{t,r} \in [0,1]$. It commits the forecast before observing $o(a_t)$. This separates the assay policy, the response predictor, and the final-action rule. The final action can be unassayed. Assays cost one budget unit; forecasts and final submission do not.

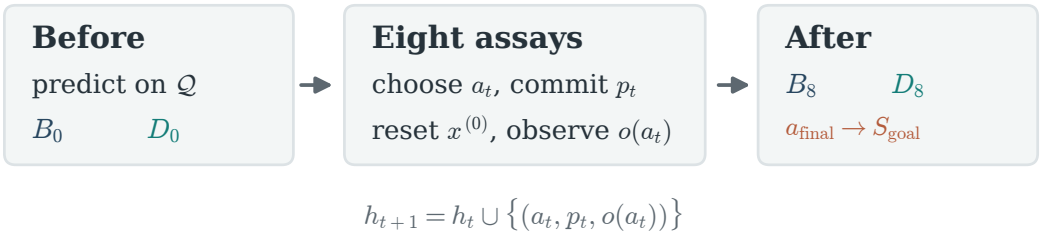


Figure 2: One episode. The evaluator scores the same ten held-out actions $\mathcal{Q}$ before and after eight assays. Probe forecasts come from copies of the procedure state and do not enter its history. Each assay starts from $x^{(0)}$; the final action is scored only for goal success.

3 Evaluation

In simulation, before each episode the evaluator selects ten legal actions $\mathcal{Q} \subset \mathcal{A}$ and excludes them from assays. Copies of the procedure before the first and after the eighth assay forecast all $10|R|$ responses. The evaluator discards each copy after scoring; probe requests never affect the acting procedure or reveal outcomes. Probes are fixed across procedures on the same task. A final action may be a probe because its outcome remains hidden until the episode ends.

Prediction error. For checkpoint $z \in \{0, 8\}$, the Brier score is the mean squared error of the predicted activation probabilities on the probes:

$$B_z = \frac{1}{10|R|} \sum_{a \in \mathcal{Q}} \sum_{r \in R} \left(p_{a,r}^{(z)} - o_r(a)\right)^2, \qquad \Delta B = B_0 - B_8. \tag{3}$$

B_z lies in $[0,1]$; positive ΔB means that forecasts improved during the eight assays. This score measures prediction on the fixed probe set. Error on executed assays answers a different question because the assay policy determines which actions are observed.

Prediction error can be low when a predictor gives the same forecast for every action. Let $q_r = 10^{-1} \sum_{a \in \mathcal{Q}} o_r(a)$ be the activation rate of readout r across probes. For example, always predicting a response that is common in the probe set can lower error without distinguishing interventions. The best constant forecast for that readout is q_r, with score

$$B_{\text{const}} = \frac{1}{|R|} \sum_{r \in R} q_r(1 - q_r), \qquad E_z = B_z - B_{\text{const}}. \tag{4}$$

B_{const} uses held-out outcomes and is an oracle reference, not a feasible procedure score. E_z is excess Brier relative to that reference. Because it subtracts the same quantity for every procedure on a task, it cannot change a paired ranking or establish action dependence.

Intervention permutation gap. To test action dependence, permute each readout's ten forecasts across probe actions while leaving their values unchanged. The *permutation gap* is the expected increase in Brier error caused by this reassignment:

$$D_z = \mathbb{E}_\pi[B_z(p_{\pi(a),r}, o_r(a))] - B_z = \frac{2}{|R|} \sum_{r \in R} \left(\overline{p_r o_r} - \overline{p}_r q_r\right). \tag{5}$$

Bars average over $\mathcal{Q}$. A constant predictor has $D_z = 0$; positive D_z means forecasts match their assigned actions better than randomly reassigned actions. We report D_0, D_8, and $\Delta D = D_8 - D_0$

to separate initial alignment, final alignment, and change. We also divide each task's D_8 by its perfect-forecast limit $2B_{\text{const}}$ when this limit is nonzero. A readout that does not vary across probes contributes zero. The gap must be read alongside prediction error: a larger gap alone does not establish more accurate probabilities or a recovered mechanism (Appendix B). Brier-score decompositions likewise distinguish marginal accuracy from resolution [Murphy, 1973].

The primary outcomes are S_{goal}, B_8, and D_8; ΔB and ΔD measure change during an episode. We also record the first assay that reaches the target. Episodes without an observed hit by assay eight are censored at eight. Goal success scores a chosen action; probe prediction scores forecasts over ten held-out actions.

Tasks and uncertainty. We construct 232 tasks from 39 published Boolean models with 9–30 nodes. Each task uses a seeded initial state, $H = 3$, four readouts, and a reachable target on one or two readouts; target nodes cannot be clamped. The ten probes are sampled before any procedure run. Appendix C specifies the model source, task filters, and selection rules.

Comparisons pair procedures on the same task and probes. We average tasks within each network, then average networks; 95% intervals resample networks because tasks from one network share transition rules. The runs of explicit rules change one component at a time where stated. The LLM prompt runs compare complete prompting procedures, so their differences cannot be assigned to an internal representation.

4 Results

4.1 Predictor and final-choice comparisons

We compare six combinations of explicit selection and prediction rules on all 232 tasks. *Random assays* sample untried interventions uniformly. *Graph-ranked assays* prioritize interventions whose signed paths suggest an effect on the target. *Uncertainty-selected assays* choose the highest ridge-prediction variance among graph-ranked candidates. None can inspect the hidden Boolean update rules.

A *constant forecast* assigns probability $1/2$ to every readout. A *mean forecast* uses the observed activation frequency of each readout, independent of the proposed intervention. *Ridge regression* fits an additive response predictor from the revealed assays, with a penalty on its coefficients. The *best-observed* final rule submits the most successful assayed intervention; the *predicted-best* rule searches the allowed interventions using the predictor. Table 1 names each combination by these operations. All rows use best-observed unless predicted-best is stated. Appendix C gives the exact rules and their parameters.

Procedure	$B_8 \downarrow$	$D_8 \uparrow$	$D_8/(2B_{\text{const}})$ (%)	Goal (%) $\uparrow$
Random assays + constant	.250	.000	0.0	54.7
Random assays + mean	**.174**	.000	0.0	54.7
Graph-ranked assays + mean	.283	.000	0.0	61.6
Graph-ranked assays + ridge	.227	.021	7.5	61.6
Graph-ranked + ridge; predicted-best	.227	.021	7.5	56.6
Uncertainty + ridge; predicted-best	.208	**.024**	**8.8**	**66.0**

Table 1: Outcomes after eight assays on 232 tasks from 39 networks. Blue denotes prediction error, teal intervention permutation gap, and rust target attainment. Means weight networks equally. The normalized D_8 averages task ratios on the 228 tasks with $B_{\text{const}} > 0$. Predicted-best searches the allowed interventions using the predictor; other rows submit the best assayed intervention. Bold marks the best mean in each column, not statistical significance.

The random-assay mean predictor has the lowest Brier score, .174, despite giving the same forecast for every action on a readout; its D_8 is therefore zero. The graph-ranked mean predictor rises from .250 before assays to .283 afterward, showing that a mean estimated from selected assays can miss probe-set activation rates (Figure 3). On 84.1% of task–readout pairs, probe responses vary across actions, so permutation gap is measurable in most cases. Directly clamped readouts account for 6.4% of probe–readout pairs; excluding them leaves $D_8 = .018$ for graph-ranked ridge and .020 for uncertainty-selected ridge.

With graph-ranked assays and the best-observed final rule held fixed, replacing the mean with ridge lowers B_8 by .056 (paired 95% network-bootstrap interval $[-.064, -.049]$) and raises D_8 by .021 $[.018, .024]$. Both procedures have 61.6% goal success because they submit the same actions (Figure 4). Perfect forecasts would have $D_8 = 2B_{\text{const}} = .287$, so the measured permutation gap remains small relative to this task-specific limit. Holding ridge forecasts and assays fixed but changing the final rule to predicted-best reduces goal success by 5.0 percentage points $[-9.4, -.7]$. Uncertainty-selected ridge reaches .208 Brier and 66.0% goal success under a different assay policy. By assay four, observed target-hit rates are 34.9% for random, 47.9% for graph-ranked, and 54.4% for uncertainty-selected ridge (Figure 3).

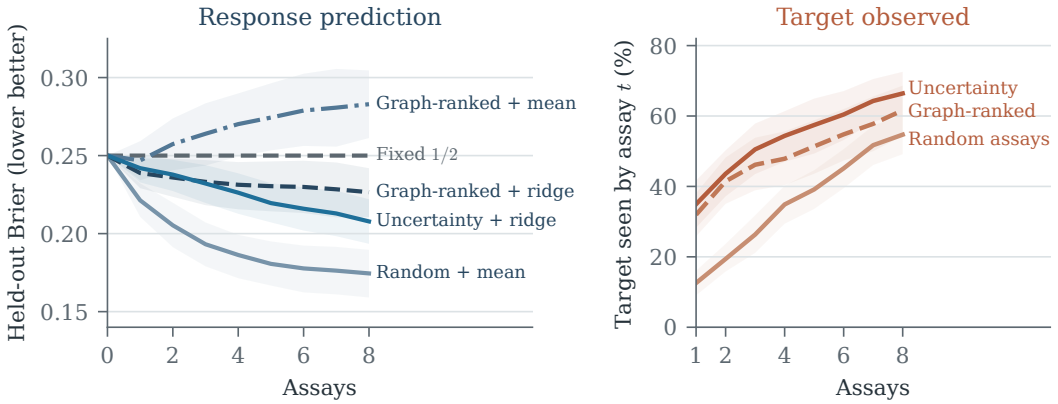


Figure 3: Outcomes through the eight-assay budget. Left: prediction error on the same held-out actions after each assay. Right: fraction of tasks with an observed target hit by assay t; episodes with no hit remain censored. Lines weight networks equally; shaded bands are 95% network-bootstrap intervals. Graph/ridge rows with different final rules share the same curve.

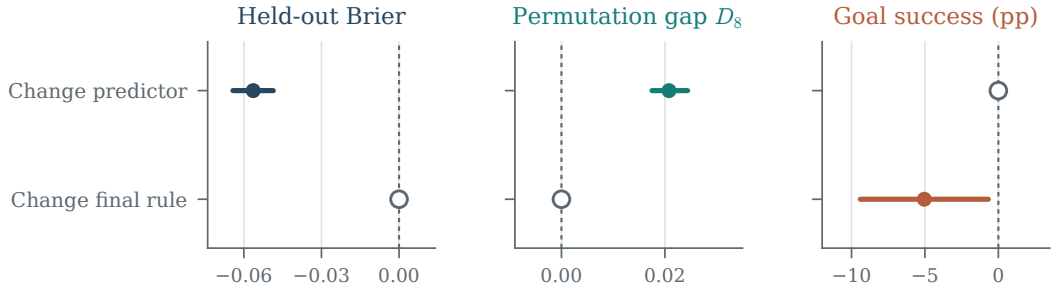


Figure 4: Paired component changes on the same tasks. Top row: replace the action-independent mean with ridge while holding graph-ranked assays and the best-observed final rule fixed. Bottom row: replace the final rule while holding ridge forecasts and assays fixed. Points are mean changes, lines are 95% network-bootstrap intervals; open circles at zero are exact equalities. Goal changes are percentage points.

4.2 LLM note comparisons

We select one task per network by a fixed hash, giving 39 paired tasks with biological node names. A second hash selects 20 of them for a paired name-anonymization control. Both prompts use model ID `openai/gpt-6-sol`, temperature zero, and a 1,024-token response limit. Each episode makes eleven calls: one initial probe forecast, eight assay decisions with forecasts, one final probe forecast, and one final action. Probe calls are stateless copies. Prompts, selection rules, response schema, and spending cap were fixed after a smoke test on a task outside the study (Appendix F).

The *No note* condition supplies the network description and all past interventions, forecasts, and observations. The *Hypothesis note* condition supplies the same information and the model's previous note, and asks it to revise that hypothesis when a response contradicts a prediction. Both conditions

choose assays, forecast before feedback, and choose a final intervention. They may therefore acquire different histories: this comparison tests the complete prompting procedure, not the note alone.

A label control replaces biological node names with opaque IDs, while preserving graph structure and the responses to each intervention. Tables distinguish *biological names* from *opaque IDs*; these labels describe what the LLM sees, not a different biological system.

Procedure on same 39 tasks	$B_8 \downarrow$	$D_8 \uparrow$	$D_8/(2B_{\text{const}})$ (%)	Goal (%)
Random assays + constant	.250	.000	0.0	56.4
Random assays + mean	.172	.000	0.0	56.4
Graph-ranked assays + mean	.282	.000	0.0	59.0
Graph-ranked assays + ridge	.225	.022	7.5	59.0
Graph-ranked + ridge; predicted-best	.225	.022	7.5	53.8
Uncertainty + ridge; predicted-best	.205	.024	8.4	69.2
LLM: No note	.174	.108	37.6	66.7
LLM: Hypothesis note	**.140**	**.140**	**47.7**	**74.4**

Table 2: Mean/ridge predictors and LLM prompts on the same 39 named tasks and held-out probes. Each row averages one task per network. Normalized permutation gap averages per-task ratios over 38 tasks with $B_{\text{const}} > 0$; the remaining task has no variable probe response. The LLM prompt rows differ in the full prompting procedure. Bold marks column extrema, not statistical significance.

On the same 39 tasks, the uncertainty-selected ridge predictor has $D_8 = .024$, compared with .108 for the LLM without a note and .140 for the LLM with a hypothesis note (Table 2). The paired gaps relative to ridge are $+.083$ $[.054, .113]$ and $+.115$ $[.088, .142]$; differences in target success remain uncertain. The corresponding normalized gaps are 8.4%, 37.6%, and 47.7%. These comparisons change the predictor and the assay-selection procedure together; they do not isolate the effect of a note.

Both prompts improve prediction during the eight assays (Figure 5). On the 39 named tasks, Hypothesis note minus No note is $+.032$ $[.007, .059]$ for final permutation gap D_8 (Figure 6). D_8 is one of three primary outcomes and we apply no multiplicity correction, so we treat this effect as suggestive. The difference in change, ΔD, is $+.030$ $[-.00003, .059]$, an interval that only just includes zero. The B_8 difference is $-.034$ $[-.069, .011]$, the ΔB difference is $+.047$ $[-.009, .098]$, and goal success differs by $+7.7$ percentage points $[-2.6, 20.5]$. Under anonymized labels the prompt difference is $+.013$ $[-.013, .043]$ for D_8 and -15.0 percentage points $[-30.0, 0.0]$ for goal success, so the named-task advantage is not reproduced without names. Because the prompts differ in hypothesis text and may also change action formatting or assay choice, this is an effect of the full prompting procedure.

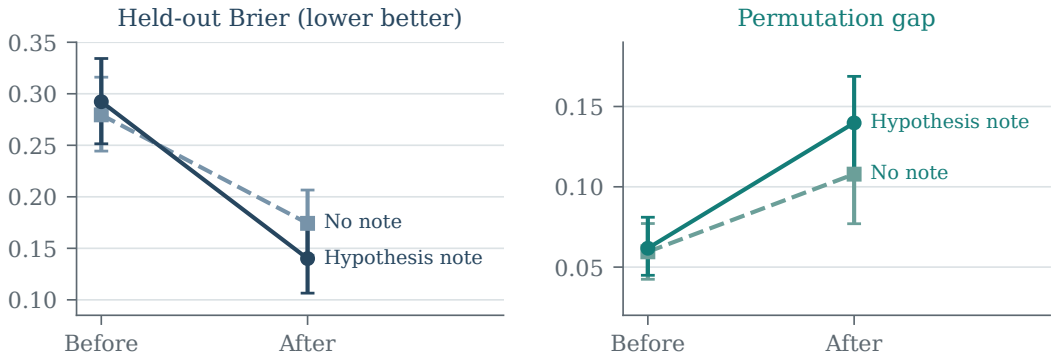


Figure 5: Named-node study (39 paired tasks). Points are means before and after eight assays; bars are 95% network-bootstrap intervals for each mean. Figure 6 tests differences between prompts on paired tasks.

On the matched 20 tasks, the label effect (anonymous minus named) is $-.048$ $[-.085, -.010]$ under the Hypothesis note prompt and $-.021$ $[-.060, .016]$ under No note (Figure 7). Their difference is $-.027$ $[-.076, .023]$, so we do not claim the label effect is prompt-specific. The corresponding

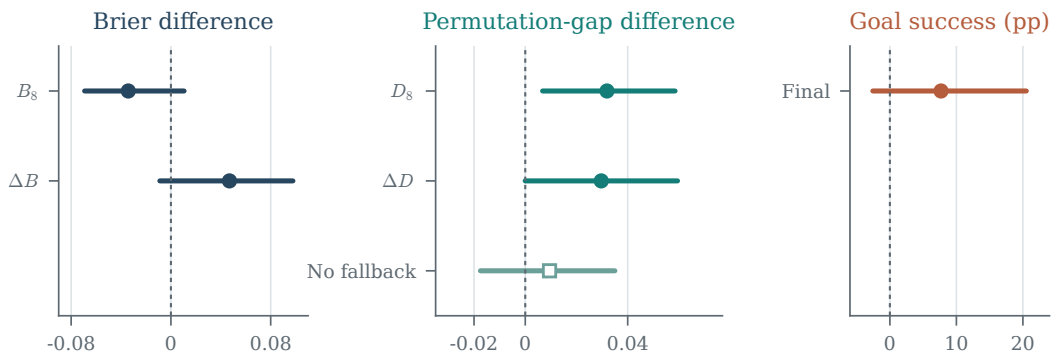


Figure 6: Hypothesis note minus No note effects on the same 39 named tasks. Points are paired means; lines are 95% network-bootstrap intervals. Dashed lines mark zero. Lower B_8 and higher ΔB, D_8, ΔD, and goal success are favorable. The open-square D_8 row uses only the 20 task pairs with no assay fallback in either condition; it is a selected-subset check.

Hypothesis note changes in D_0 ($-.019$ $[-.045, .003]$), ΔD ($-.029$ $[-.078, .021]$), Brier, and goal success have intervals spanning zero. Label replacement does not identify which facts, if any, the names supplied.

Procedure on same 20 tasks	$B_8 \downarrow$	$D_8 \uparrow$	$D_8/(2B_{\text{const}})$ (%)	Goal (%)
Random assays + constant	.250	.000	0.0	60.0
Random assays + mean	.192	.000	0.0	60.0
Graph-ranked assays + mean	.281	.000	0.0	50.0
Graph-ranked assays + ridge	.223	.027	8.2	50.0
Graph-ranked + ridge; predicted-best	.223	.027	8.2	45.0
Uncertainty + ridge; predicted-best	.208	.030	9.2	60.0
Biological names: No note	.179	.112	36.2	50.0
Biological names: Hypothesis note	**.142**	**.152**	**48.0**	60.0
Opaque IDs: No note	.196	.091	27.4	**65.0**
Opaque IDs: Hypothesis note	.174	.105	33.0	50.0

Table 3: Mean and ridge predictors and both LLM prompts with named or opaque labels on the same 20 tasks. Normalized D_8 is the mean of 20 per-task ratios. The explicit selection rules use graph structure and node indices, so relabeling does not change their rows. Bold marks column extrema, not statistical significance.

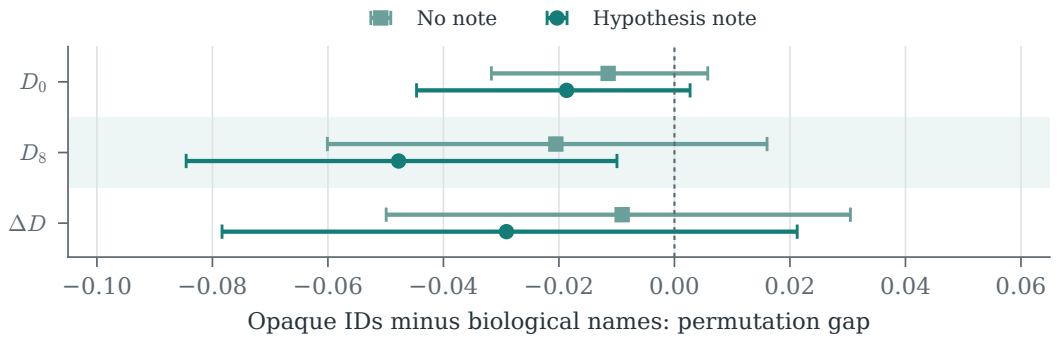


Figure 7: Paired anonymous-minus-named effects on D_0, D_8, and ΔD for the same 20 tasks. Bars are 95% network-bootstrap intervals. Only the Hypothesis note prompt's D_8 interval excludes zero.

With biological names visible, invalid actions triggered the preset fallback on 42 of 312 assays under No note and 27 of 312 under Hypothesis note. In the 20 named task pairs with no fallback in either condition, the D_8 difference is $+.009$ $[-.018, .035]$. This selected subset cannot measure how much

of the full difference came from formatting failures. No probe forecast was malformed. Appendix D reports all failures, token counts, and paired intervals. The original paired study used 1,298 calls; estimated cost including smoke tests was $4.83.

Post hoc note control. We then added an *Observation note* condition on the same 39 tasks: the LLM summarizes observations without being asked to form or revise a hypothesis (Appendix F). Its mean $D_8 = .119$, mean per-task normalized D_8 is 42.3%, $B_8 = .159$, and goal success is 71.8%; it uses 1,132 output tokens per episode, versus 643 for No note and 1,046 for Hypothesis note. Relative to No note, Observation note minus No note is $+.011$ $[-.016, .036]$ for D_8, $-.015$ $[-.058, .037]$ for B_8, and $+5.1$ percentage points $[-12.8, 23.1]$ for goal success. Relative to Hypothesis note, the differences are $-.021$ $[-.045, .002]$, $+.019$ $[-.016, .058]$, and -2.6 percentage points $[-17.9, 12.8]$. All intervals cross zero. This single post hoc run does not resolve whether hypothesis wording, longer notes, or other prompt effects explain the original difference.

4.3 Association between prediction and search outcomes

Across the 232 simulation tasks, the mean within-procedure Spearman correlation between D_8 and binary goal success is $\rho = .108$ $[-.010, .227]$ for the three ridge procedures with variable D_8. The uncertainty-selected ridge procedure alone has $\rho = .212$ $[.085, .334]$; the fixed and mean predictors have constant D_8, so rank correlation is undefined for them. On the 39 named tasks, the mean correlation across the No note and Hypothesis note conditions is $\rho = -.116$ $[-.380, .159]$; pooling their 78 episodes gives $\rho = -.089$ $[-.351, .177]$. Intervals resample networks and repeat the rank calculation. Thus success and permutation gap do not move together consistently across these procedures; cross-task correlations do not isolate a causal relationship.

4.4 Replay of measured perturbations

Measurements. We replay published outcomes from six human T-cell cytokine screens [Schmidt et al., 2022] and single/double CRISPRa perturbations in K562 cells [Norman et al., 2019]. In each T-cell screen, the procedure sees gene identities, perturbation modality, cell context, and public gene-family annotations. It searches 128 candidate genes for the largest measured effect; 32 different genes are reserved for prediction. The effect is $\log_2$ guide enrichment between cytokine-high and cytokine-low sorting bins, not cytokine concentration. CRISPRa denotes gene activation and CRISPRi gene interference. Screens remain separate by cell type, cytokine, and perturbation modality.

For K562 cells, every predictor receives measured responses to the constituent single perturbations. It then forecasts measured combinations on an eight-gene expression panel selected using singles only. Of 126 eligible combinations, each split reserves 26 for prediction and permits eight assays from the other 100. This experiment evaluates combination prediction; it has no search objective. Both datasets use five fixed splits. Appendix G specifies preprocessing and eligibility.

Comparisons. Unlike the simulation prompt comparison, the primary LLM comparison here fixes the eight assay identities and observations, then changes only the note instruction: *Observation note* summarizes what was measured; *Hypothesis note* proposes and revises an explanation of response patterns. Both use the same 600-character note limit and output-token ceiling. Mean and ridge predictors provide references; the sum of the two single responses is an additional K562 reference. We measure mean squared error (MSE) and the permutation gap before and after feedback. Separate runs let the LLM choose assays or shuffle the feedback across intervention identities. Split ranges describe evaluation variability, not independent biological replicates.

The matched-history results do not show a consistent final-MSE advantage for hypothesis notes across screens (Table 4). Initial and final errors are reported separately in Appendix H; a difference in final error alone does not establish a larger improvement from feedback.

When the LLM chooses its own assays, search rankings also depend on the screen: no procedure has the lowest mean regret (best candidate effect minus chosen effect) in all six T-cell screens. These comparisons change both the acquired history and the final choice, so they measure the complete search procedure rather than isolate prediction learning.

Screen	n	Final MSE difference	Split range
Norman K562	5	-0.008	$[-0.061, +0.054]$
P. CRISPRi CD8 IFNG	5	$+0.003$	$[-0.003, +0.011]$
P. CRISPRa CD4 IL2	5	-0.006	$[-0.013, +0.000]$
P. CRISPRa CD8 IFNG	5	$+0.000$	$[-0.003, +0.003]$
P. CRISPRi CD4 IL2	5	$+0.006$	$[+0.001, +0.012]$
S. CRISPRa CD4 IFNG	5	-0.001	$[-0.009, +0.003]$
S. CRISPRa CD4 IL2	5	$+0.001$	$[-0.003, +0.009]$

Table 4: Hypothesis note minus Observation note on identical assay histories. Negative MSE differences favor Hypothesis note. T-cell rows use squared $\log_2$ enrichment; Norman uses squared changes in $\log_2$(1+CPM) on its selected panel. Ranges show all available split differences, not biological confidence intervals. P. and S. denote primary and supplementary source screens.

On the Norman panel, final MSE is 0.735 for Observation note and 0.727 for Hypothesis note, compared with 0.918 for the sum-of-singles reference. Response error and additive-residual error are algebraically equal; the permutation gap of the residuals is reported separately rather than counted as a second error result. Figures 8–10 show prediction, search, and inference comparisons.

The pilot completed 105/105 paired blocks with $28.06 accounted against the budget, including format checks and conservative reservations for interrupted calls. Original study API latency has median 3.06s and 95th percentile 6.44s; CPU decision latency has median 0.05ms. These measurements include six-way episode concurrency and exclude cache replay. We retain 49 malformed responses and their bounded repairs in the logs. This pilot supports claims about these measured screens and the disclosed panel, not biological mechanism recovery, prospective discovery, or contamination-free generalization.

5 Discussion

The controlled simulator comparisons show why prediction and search outcomes should be reported separately. Replacing the mean predictor with ridge improves held-out predictions at fixed final choices; replacing the final-choice rule changes success at fixed forecasts. Target success alone cannot establish learning to predict untried interventions.

Hypothesis notes increase final permutation gap in the original simulation comparison, but learning-gain and success differences remain uncertain, and sensitivity checks weaken the effect. Measured-data replay on fixed histories finds no consistent final-error advantage over observation notes. These results do not establish a general benefit from hypothesis notes.

The evidence is limited to the tested tasks, response horizons, and measured gene panel. Ten simulation probes cannot identify unique update rules, and the Boolean networks are not validated substitutes for cells [Park et al., 2025]. Replaying historical measurements does not establish prospective discovery; public outcomes may also overlap LLM training data. We evaluate one LLM with one temperature-zero run per condition and task, without estimating repeated-run variability. In the original simulation comparison, Hypothesis note also produces more output tokens than No note (1,046 versus 643 per episode), so its effect cannot be attributed solely to maintaining a hypothesis.

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A Counterexample and metric checks

Set $F_u(x) = u$, $F_v(x) = v$, and $F_y(x) = u \wedge \neg v$, with $H = 1$. No clamp gives $y_1 = 0$. Figure 1 covers all four input settings. The target is $y_1 = 1$ and y cannot be clamped. Both procedures attain $S_{\text{goal}} = 1$; on reserved $(1, 1)$ their Brier losses are $(0-0)^2 = 0$ and $(1-0)^2 = 1$. The construction proves a logical distinction, not its prevalence.

Equation (3) averages exactly $10|R|$ terms, each in $[0, 1]$. Hence $B_z \in [0, 1]$ and $\Delta B \in [-1, 1]$. Equation (2) is binary. Neither metric includes an intervention cost; report the number of assays and intervention cardinality separately.

B Metric derivation

For one readout, write $p_a = p_{a,r}$ and $y_a = o_r(a)$. Averaging the squared loss after a uniform random permutation π of the ten forecast values gives $\mathbb{E}_\pi \overline{(p_{\pi(a)} - y_a)^2} = \overline{p^2} - 2\overline{py} + \overline{y}$. Subtracting the original loss $\overline{p^2} - 2\overline{py} + \overline{y}$ yields $2(\overline{py} - \overline{p}\,\overline{y})$. Average this identity over readouts to obtain Eq. (5). Constant forecasts and constant outcomes each make the covariance zero. A perfect binary forecast on a readout of prevalence q has permutation gap $2q(1-q)$; reversing its bits gives the negative of this value. The score lies in $[-.5, .5]$. The oracle constant minimizes squared loss over forecasts that use only the readout identity, since $\overline{(c-y)^2} = (c-q)^2 + q(1-q)$. For the direct-clamp check, we recompute the covariance for each readout on $\{a \in \mathcal{Q} : r \notin \mathrm{dom}(a)\}$, then average over readouts; a readout with fewer than two remaining probes contributes zero.

C Instance construction and comparisons

Source and eligibility. The source is `Benchmarks/models` from boolmore commit `a757a6d` (full hash in the accompanying manifest). Of its 40 models with at most 30 nodes, we use the 39 with 9–30 nodes. The repository includes each model, a checksum manifest, and its source license. The parser accepts only identifiers, constants, negation, conjunction, and disjunction. For seeds 0, 1, and 2, a model-specific SHA-256 seed selects the initial bit vector. All assays use $H = 3$ synchronous updates. Enumerate every nonempty clamp of one or two nodes, including clamps to their initial value. For each node, compute activation frequency over all such interventions; the four nodes with largest $p(1-p)$ form R.

For each one- or two-node subset of R, set target bits opposite their unperturbed H-step values. Exclude target nodes from legal clamps. Retain targets achieved by 1–35% of legal actions and with at least 18 legal actions. Rank eligible targets by larger target set, then proximity of reachable fraction to 10%, then node ID; choose at most two per model and seed, preferring a new target node. This gives 232 tasks on all 39 models. The oracle is successful by construction and is not an independently measured result.

Probe and target choice. Generate interventions in node-index and value order. Sample ten legal probes without replacement using a hash of the model seed and target, before any procedure run. The same probes and initial state apply to all procedures on a task; probes are absent from the assay menu. A final action may be any legal intervention. To provide a signed graph, enumerate each rule's input truth table, mark a dependency activating or inhibiting if its effect has one sign, and omit mixed or ineffective dependencies. The graph is visible; truth tables and rules are not.

Action baselines. Random assay selection draws eight distinct interventions uniformly from the non-probe menu. Graph ranking scores an intervention by summing, over target nodes, the best $1/(1+d)$ for a directed signed path of length d whose sign matches the desired change relative to the initial bits; it subtracts 0.01 for a second clamp. Ties prefer fewer clamps and canonical order. Graph-ranked mean and both graph-ranked ridge variants run the same eight actions. Uncertainty-selected ridge takes the top 40 unused graph-ranked actions and chooses the one of greatest ridge predictive variance, with graph rank as tie breaker. Best-observed procedures submit the observed action with fewest target-bit errors, then fewest clamps; predicted-best procedures submit the action of maximum predicted independent-bit target probability over all legal actions. The oracle evaluates all legal actions using F and selects a successful one when available.

Prediction baselines. The fixed $1/2$ forecast has Brier score $1/4$ on every binary response. Mean uses the within-episode readout count with a Beta$(1,1)$ prior, regardless of action. Ridge fits readout probabilities with kernel $k(a,b) = 1 + |a \cap b|$ on node-value assignments, penalty 2, and prior mean $1/2$: $\hat{p}_r(a) = \mathrm{clip}_{[0,1]}\{1/2 + k_a^\top (K + 2I)^{-1}(y_r - 1/2)\}$. Its acquisition variance is $k(a,a) - k_a^\top (K + 2I)^{-1} k_a$. All predictions are committed before the corresponding assay and scored on all four readouts.

Comparison unit. The evaluation of explicit rules contains 1,392 episodes: 232 tasks times six procedures. Average task scores within each model, then average the 39 model means. For paired

comparisons, first subtract procedures on each identical task, then average differences by model. Intervals are percentile intervals from 2,000 bootstrap resamples of models with fixed seed 20260924. The accompanying JSONL records each action, prediction, observation, probe forecast and outcome, and final choice; the task manifest records probes and checksums. These intervals describe variation across selected models and seeds, conditional on this task construction.

D Additional outcome detail

Time to first observed hit. The entries below give the percentage of episodes with a hit by the stated assay number. An episode without a hit by eight is censored. Rows using random, graph-ranked, or uncertainty-selected assays average tasks within model, then models; LLM rows have one task per model. The anonymous rows use a 20-model subset of the named run. Bold marks the largest percentage within each separated task cohort, including ties.

Procedure	By 1	By 2	By 4	By 8
Random assays + mean	12.6	19.4	34.9	54.7
Graph-ranked assays + mean	32.0	41.5	47.9	61.6
Uncertainty + ridge	**35.0**	**43.6**	**54.4**	**66.4**
Biological names: No note	**43.6**	48.7	**61.5**	**74.4**
Biological names: Hypothesis note	**43.6**	**53.8**	**61.5**	**74.4**
Opaque IDs: No note	**30.0**	**35.0**	**50.0**	**55.0**
Opaque IDs: Hypothesis note	20.0	25.0	45.0	**55.0**

Paired LLM prompt intervals. All brackets are 95% bootstrap intervals over networks. On 39 named tasks, Hypothesis note minus No note is $+.013$ $[-.015, .044]$ for B_0, $-.034$ $[-.069, .011]$ for B_8, $+.047$ $[-.009, .098]$ for ΔB, $+.002$ $[-.010, .015]$ for D_0, $+.032$ $[.007, .059]$ for D_8, and $+.02967$ $[-.000033, .05946]$ for ΔD. Goal success differs by $+.077$ $[-.026, .205]$. The oracle reference $B_{\text{const}} = .145$ on these tasks; $E_8 = .029$ for No note and $-.005$ for Hypothesis note. Subtracting the same reference means the paired E_8 and B_8 differences are identical.

On 20 anonymous tasks, Hypothesis note minus No note is $+.007$ $[-.022, .039]$ for B_0, $-.022$ $[-.087, .041]$ for B_8, $+.029$ $[-.043, .104]$ for ΔB, $+.001$ $[-.010, .011]$ for D_0, $+.013$ $[-.013, .043]$ for D_8, and $+.013$ $[-.018, .049]$ for ΔD. Goal success differs by $-.150$ $[-.300, .000]$. On the matched 20 tasks, anonymous minus named under the No note prompt gives $-.011$ $[-.032, .006]$ for D_0, $-.021$ $[-.060, .016]$ for D_8, and $-.009$ $[-.050, .030]$ for ΔD. Under the Hypothesis note prompt these are $-.019$ $[-.045, .003]$, $-.048$ $[-.085, -.010]$, and $-.029$ $[-.078, .021]$. The corresponding B_8 effects are $+.017$ $[-.031, .077]$ and $+.032$ $[-.017, .085]$. These are effects of label replacement under two prompting procedures, not estimates of biological knowledge alone.

Response validation and cost.

Condition	Tasks	Invalid assays	Invalid final	Input tokens	Output tokens
Biological names: No note	39	42/312	1	14,422	643
Biological names: Hypothesis note	39	27/312	0	15,138	1,046
Opaque IDs: No note	20	21/160	2	14,502	657
Opaque IDs: Hypothesis note	20	12/160	1	15,255	1,112

Tokens are means per eleven-call episode. Invalid assay actions use the first legal unused action and four probabilities of $1/2$; invalid final actions use the best observed action. No probe response failed JSON validation. Restricting to task pairs with no assay-action fallback in either named condition leaves 20 tasks: Hypothesis note minus No note $D_8 = +.009$ $[-.018, .035]$ and $\Delta D = +.007$ $[-.022, .033]$. This post-selection sensitivity check has less precision and does not identify a formatting effect. The 118 original study episodes used 1,298 calls (118×11); the log also contains 42 smoke-test calls, for 1,340 in total before the post hoc control. Estimated original cost was $4.8333 against the $50 cap. SDK version was 2.54.0; all logged original study calls report zero reasoning tokens.

E Interface and information access

The evaluator exposes `inspect_network()`, `predict(a,p)`, `experiment(a)`, and `submit(a)`. Inspection returns the graph, initial state, horizon, readouts, target, assay menu, and final actions. Prediction commits one probability per readout without spending an assay. Experiment requires a matching commitment, returns the response, and spends one assay. Submission scores the final action. Invalid or repeated assays, forbidden probes, target clamps, and actions after eight assays reveal no response and spend no budget.

Probe evaluation clones the procedure at the checkpoint, asks for ten forecast vectors, and discards the clone. The acting state receives no probe outcome. Logs retain the task manifest, model checksum, procedure configuration, prompts, actions, predictions, observations, probe forecasts, and final action. Hidden rules remain evaluator-only.

F LLM prompt execution

Selection and pairing. For each of the 39 models, generate the existing seed-0, seed-1, and seed-2 tasks and select the task minimizing a SHA-256 hash of a fixed public salt and task ID. A separate hash of the model name determines study order; a third selects the first 20 models for anonymization. The accompanying selection manifest gives the exact salt and IDs. We run each named No note/Hypothesis note pair before its next named task, then each anonymous pair in hash order. A $50 cap, including smoke-test calls, stops the run before a request that might exceed it; only completed pairs enter analysis. A different, nonselected task supplied the API smoke test and contributed no study score. The anonymous map is a model-specific seeded permutation from node indices to `node_00`, `node_01`, and so on. No model title is placed in either procedure prompt. The source rules remain available only to the evaluator.

Request format. The static system instruction reads: "You are an experimental agent studying a hidden deterministic Boolean system. Each assay resets to the disclosed initial state. The signed graph may be incomplete. You do not know the update rules. Return one valid JSON object only, with no Markdown. Probability arrays follow the stated readout order. Be concise." The Hypothesis note condition adds: "Maintain a short, revisable response hypothesis in the note field. Revise it when observations disagree with predictions." Every stateless request includes the phase, network view, and complete prior action–prediction–observation triples. The view specifies node identifiers, initial bits, horizon, readouts, target, signed edges, eligible clamp nodes, ten forbidden assay actions, budget, and legal-action rule. The Hypothesis note condition additionally includes its prior hypothesis.

For an assay, the response schema is `{"action":[["node",0]],"p":[0.5,0.5,0.5,0.5]}`; the Hypothesis note condition also returns a short `note`. A probe request supplies the ten actions and asks for `{"predictions":[[p1,p2,p3,p4],...]}` in their given order; responses are discarded after scoring. A final request asks for `{"action":[["node",0]]}`. The request payloads and raw responses are logged. No probe outcome appears in any request. A malformed assay action is replaced by the first legal unused action and its forecast by four halves. A malformed forecast alone becomes four halves; malformed probe output becomes ten vectors of halves; a malformed final action becomes the best observed action. There is no retry. Failures, token usage, and cost are logged per episode. These rules preserve eleven calls per condition and are reported rather than silently discarded.

Post hoc note control. After inspecting the original results, we ran a third condition on the same 39 named tasks. It retains the No note prompt and adds this system instruction: "Use the note field to record a brief summary of what you have observed so far." Each assay returns a short `note`, and the next request carries the latest note as `previous_note`, truncated to 800 characters. It does not ask for a hypothesis or revision after prediction errors. The model, decoding settings, assay budget, probes, response validation, and eleven-call sequence match the original conditions. This control was not prespecified and has one run per task. Its 39 episodes used 429 additional calls at an estimated cost of $1.6180; all 1,769 logged calls, including the original smoke tests, cost $6.4513.

G Measured perturbation replay protocol

Question and source measurements. The pilot asks whether the two LLM prompt procedures improve predictions of untried perturbations when given the same observations. Its primary comparison is final held-out MSE, Hypothesis note minus Observation note, within each biological screen. Adaptive search and shuffled feedback are secondary comparisons. It is a retrospective replay of published measurements, not prospective wet-lab validation.

Schmidt et al. [Schmidt et al., 2022] measured guide enrichment between cytokine-high and cytokine-low sorting bins in stimulated primary human T cells. We retain the published gene-level $\log_2$ fold changes from Supplementary Table 1, archived by BioGRID ORCS. Six screens remain separate: primary CD4 IL-2 and CD8 IFN-γ screens under CRISPR activation and interference, and supplementary CD4 IL-2 and IFN-γ activation screens. The endpoint is guide enrichment, not cytokine concentration or transcript abundance. All finite gene effects with at least two guides are eligible; neither significance nor effect size determines eligibility. Each of five seeded splits samples 128 assay/final candidates and 32 disjoint probes. Direct cytokine-gene interventions remain eligible. No outcome is generated for an unmeasured action. The summary table does not expose donor-level effects; guide counts are not independent biological replicates.

Norman et al. [Norman et al., 2019] measured single and combined CRISPRa perturbations in K562 cells (GSE133344). We aggregate original filtered unique molecular identifier (UMI) counts after retaining cells with good coverage, one assigned cell, and concordant guide-identity halves. The resulting archive contains 90,634 cells across 235 perturbation/control groups. A disclosed single or evaluated combination requires at least 30 cells; both constituent singles must be available. This leaves 126 eligible combinations. For each perturbation, pooled UMI counts are normalized to counts per million (CPM), transformed by $\log_2(1 + \mathrm{CPM})$, and contrasted with the non-targeting control. This is a pooled-expression contrast, not the mean of separately normalized cells. Technical GEM-group identifiers are retained but are not biological replicate identifiers.

The eight readout genes have greatest response variance across disclosed singles only, with Ensembl ID resolving ties: HBE1, LST1, AVP, HBZ, CLC, HBA2, CHI3L2, and ALAS2. None of these readout genes is itself an evaluated perturbation target. All eligible single responses on this panel are disclosed to every predictor. Feature scaling also uses singles only. No combination outcome enters readout selection or predictor scaling. Each of five seeded splits reserves 26 combinations for probes and 100 for assays, of which eight are revealed. This tests prediction of measured combinations with known constituent singles on a selected panel, not whole-transcriptome prediction or unseen-single generalization. There is no invented scalar optimization target for this dataset.

Predictors, assay choices, and note instructions. Random assays + mean forecasts zero initially and the observed response mean after feedback. The ridge predictor is a Gaussian linear posterior mean with unit observation variance and prior precision 2. For T cells, its features are an intercept and a unit-normalized multi-hot vector of public HGNC gene-family memberships. Exact gene-symbol matches are used; unannotated genes have an intercept only. The LLM receives the same memberships. For Norman, features concatenate an intercept, the sum of the two disclosed single responses, and their absolute difference, scaled by single-response standard deviations. Ridge learns residuals around the additive single-response forecast. The unchanged additive prediction is a separate reference. Random assays + ridge fixes the random assay history; Uncertainty + ridge selects the largest posterior latent variance, with fixed-hash tie breaking. Final choices from mean/ridge procedures maximize observed T-cell enrichment.

Both LLM note conditions use `openai/gpt-6-sol`, temperature zero, no reasoning effort, a 1,536-token output limit, and a 600-character note limit. The Observation note condition records observations; the Hypothesis note condition maintains a revisable response hypothesis. Equal limits do not force equal realized token usage. Each T-cell episode has eleven logical calls: initial probe forecasts, eight assay forecasts committed before feedback, final probe forecasts, and a final action. Norman omits the final-action call. Probe calls do not update the acting note. In the primary comparison, the eight random assay identities and observed values are identical between conditions, and histories contain these observations without condition-specific earlier forecasts. Adaptive conditions choose their own assays. The shuffled-feedback diagnostic permutes the eight fixed assay outcomes among their identities before sequential disclosure, following prior feedback-ablation work [Gupta et al., 2025]; it never uses probe outcomes.

Malformed responses receive one schema-only repair request. After a second failure, the shared ridge predictor supplies forecasts from visible observations only; an invalid assay uses the first allowed action, and an invalid final choice uses the best visibly observed response. All failures remain in the analysis. Requests and responses are cached by exact request hash. Smoke-test episodes use a separate seed and are excluded. Dataset, split, prompt, metric, and executable-code hashes are frozen before scored inference; this is a local protocol freeze, not an external preregistration.

Scoring and uncertainty. For continuous measured outcomes, we report MSE and use Equation 5 with real-valued forecasts and outcomes: the resulting twice-covariance remains the expected increase in MSE after randomly permuting intervention identities. It has squared assay units and is not a proper score. Initial and final scores separate prior alignment from changes after feedback. For Norman, the additive reference defines a residual for each measured combination. Subtracting the same reference from prediction and truth leaves MSE unchanged, so response and residual MSE are not independent evidence; residual permutation gap additionally measures alignment of predicted nonadditivity. No residual is claimed to identify a molecular mechanism.

T-cell search outcomes are final measured enrichment, regret (the best measured enrichment in the fixed 128-gene pool minus the enrichment of the final choice), and the best observed effect after each assay. Different screens are not pooled into one biological effect. Five overlapping seeded splits describe protocol variability. We report all split values, means, standard deviations, and ranges, without interpreting split ranges as biological confidence intervals or bootstrapping individual cells as biological replicates. Public historical measurements may overlap language-model training data.

Compute and audit. The pilot reserves at most $5 for format checks and $45 for scored inference under a shared $50 ledger. Each paired block reserves a conservative upper bound before launch; every request records its reservation before sending, and unresolved requests retain that charge after a crash. Pricing is pinned to the verified Prime listing of $2 and $10 per million input and output tokens. Complete pairs enter paired analysis; incomplete blocks and their costs remain logged. No GPU rental or model training is used. The scored runner uses six concurrent independent episodes. Timing distinguishes original API latency, decision time, episode wall time, and cache replay. Independent audits recompute outcomes and scores, verify paired histories and action/probe isolation, inspect every request's disclosed history, and reconcile the spending ledger.

Relation to prior work.

Prior work	Established result and the pilot's narrower addition
BioDiscoveryAgent	Biological hit discovery through sequential perturbation selection. We additionally pair search outcomes with before/after held-out forecasts under matched histories.
Systema	Evaluation beyond common perturbation-response variation. Our action-identity diagnostic is used within sequential episodes, without claiming a new proper scoring rule.
GEARS	Prediction of multigene responses and genetic interactions. This pilot tests an eight-assay update with disclosed singles and an additive reference; it does not claim to outperform GEARS.
Gupta et al.	Feedback permutation and Bayesian-optimization comparisons. We reuse this control and score held-out forecasts; feedback shuffling is not new.

Our contribution is the controlled combination of prediction and search evaluations under a shared budget, not a new perturbation predictor, dataset, or biological mechanism. The novelty comparison does not establish an exhaustive priority claim.

H Measured pilot results by screen

All entries average available seeded splits. Reference rows use mean, ridge, or summed-single forecasts. Same assays fixes the LLM observations; LLM-selected lets the LLM choose assays; Shuffled feedback reassigns observed outcomes to the fixed assay identities. Standard deviations and complete split values are retained in the accompanying JSON; the splits reuse one published measurement table and are not independent biological experiments. Blue shades prediction error, teal the permutation gap, and rust search outcomes. Bold marks lowest final MSE or regret and highest final effect within each comparison group and screen, including ties at displayed precision; it does not indicate significance. The permutation gap is a diagnostic and is not ranked as a proper score.

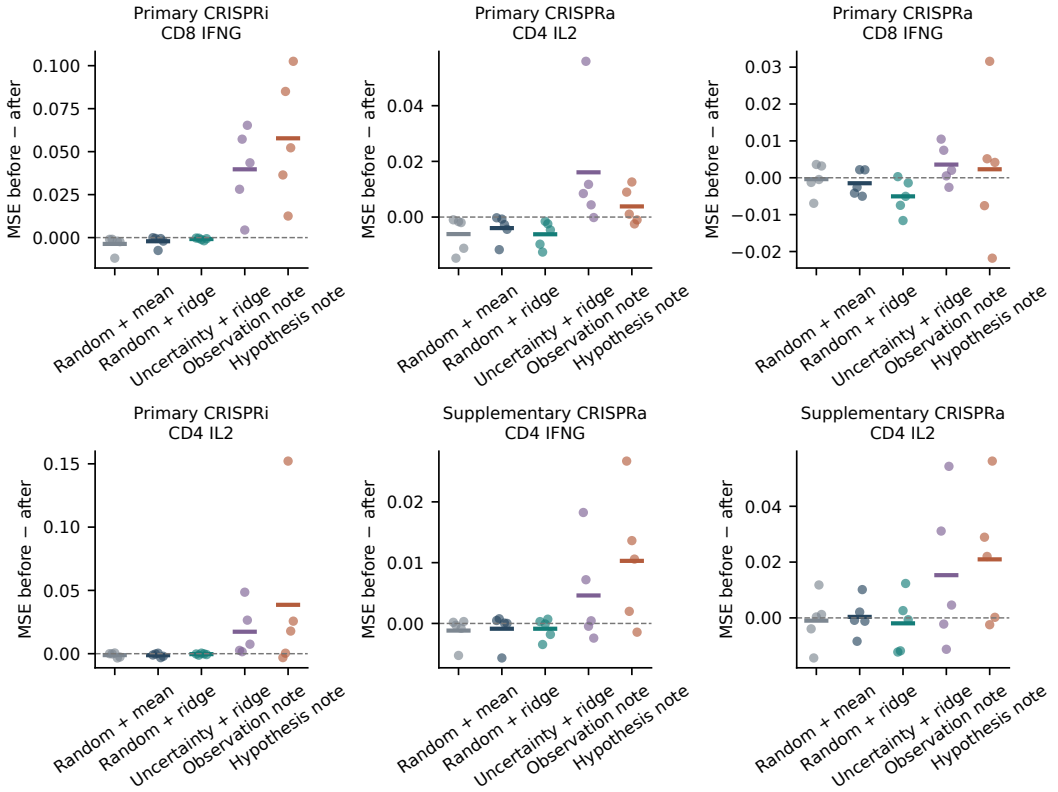


Figure 8: Held-out MSE improvement by T-cell screen, initial minus final. Positive values indicate improvement; dashed lines mark no change. Points are the five seeded split values; horizontal bars are their means. LLM conditions use identical random assay histories. Uncertainty + ridge selects different assays. Observation note and Hypothesis note are LLM prompts. Axes retain each screen's squared $\log_2$ enrichment units.

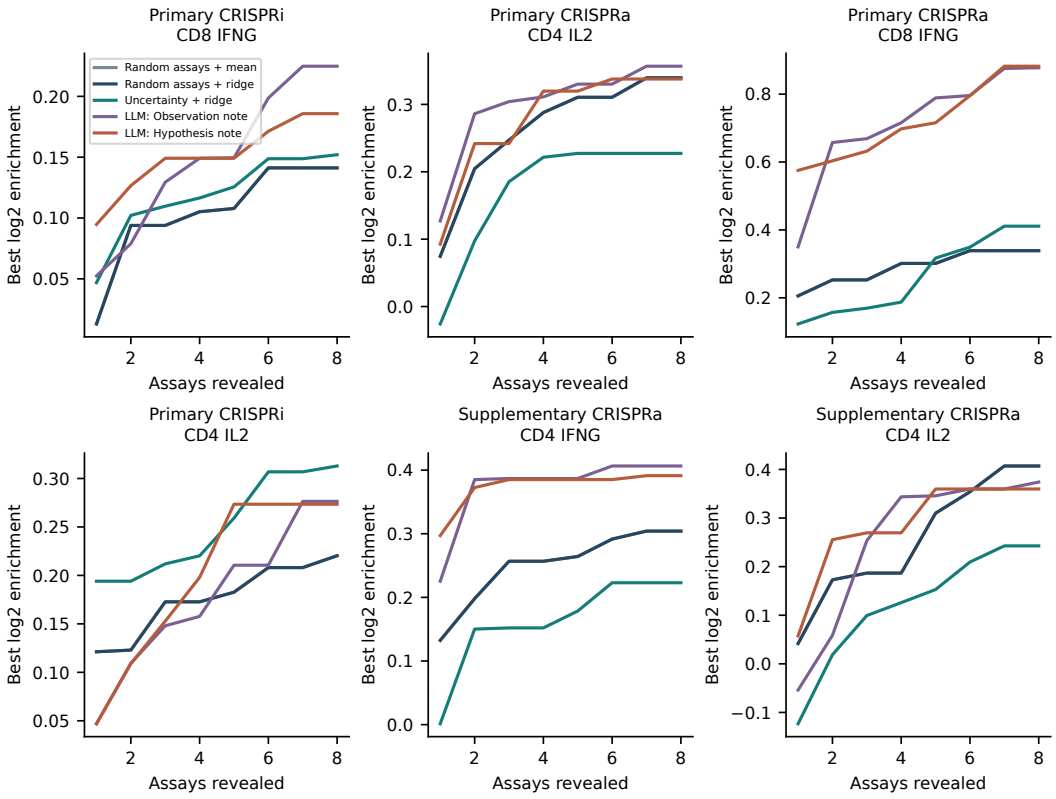


Figure 9: Best observed T-cell enrichment versus assays revealed, averaged over available seeded splits. LLM curves use adaptive acquisition. Each panel represents a distinct source screen; curves do not imply independent biological replication. Final chosen effects and regret are tabulated separately.

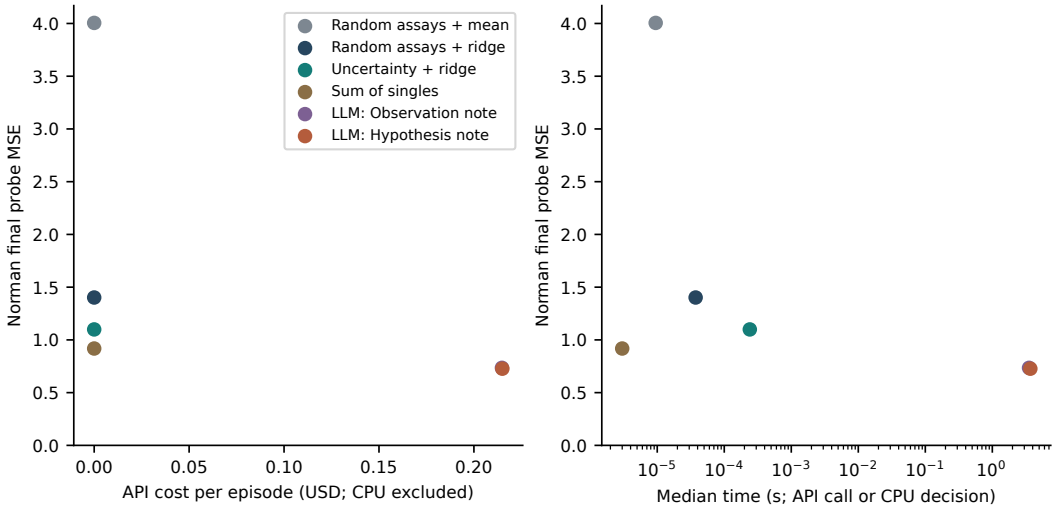


Figure 10: Norman prediction error versus API cost and time. Each point averages split-level final MSE. Mean, ridge, and sum-of-singles predictors incur zero API charges; their local compute cost is not monetized. Time is median original API-call latency for LLMs and CPU decision latency for the explicit selection rules; the operations differ, so the plot is a resource description rather than a matched-kernel speed comparison.

H.1 Norman K562

Comparison	Prediction procedure	n	MSE_0	$\text{MSE}_8 \downarrow$	D_8
LLM-selected	LLM: Hypothesis note	5	0.767	**0.831**	4.918
LLM-selected	LLM: Observation note	5	0.749	0.832	4.965
Reference	Sum of singles	5	0.918	**0.918**	6.586
Reference	Random assays + mean	5	5.142	4.005	0.000
Reference	Uncertainty + ridge	5	0.918	1.099	5.884
Reference	Random assays + ridge	5	0.918	1.402	6.037
Same assays	LLM: Hypothesis note	5	0.857	**0.727**	5.124
Same assays	LLM: Observation note	5	0.746	0.735	5.177
Shuffled feedback	LLM: Hypothesis note	5	0.764	1.044	4.225
Shuffled feedback	LLM: Observation note	5	0.763	**0.790**	4.826

Comparison	Prediction procedure	Residual permutation gap
LLM-selected	LLM: Hypothesis note	0.669
LLM-selected	LLM: Observation note	0.646
Reference	Sum of singles	0.000
Reference	Random assays + mean	1.784
Reference	Uncertainty + ridge	0.332
Reference	Random assays + ridge	0.307
Same assays	LLM: Hypothesis note	0.657
Same assays	LLM: Observation note	0.629
Shuffled feedback	LLM: Hypothesis note	0.846
Shuffled feedback	LLM: Observation note	0.718

H.2 Primary CRISPRi CD8 IFNG

Comparison	Prediction procedure	n	MSE_0	$\text{MSE}_8 \downarrow$	D_8
LLM-selected	LLM: Hypothesis note	5	0.087	**0.022**	0.001
LLM-selected	LLM: Observation note	5	0.062	0.024	-0.000
Reference	Random assays + mean	5	0.019	0.023	0.000
Reference	Uncertainty + ridge	5	0.019	**0.020**	-0.000
Reference	Random assays + ridge	5	0.019	0.021	-0.000
Same assays	LLM: Hypothesis note	5	0.084	0.026	0.000
Same assays	LLM: Observation note	5	0.062	**0.023**	-0.000
Shuffled feedback	LLM: Hypothesis note	5	0.053	0.025	-0.002
Shuffled feedback	LLM: Observation note	5	0.058	**0.024**	0.000

Comparison	Prediction procedure	Final effect ↑	Regret ↓
LLM-selected	LLM: Hypothesis note	0.186	0.258
LLM-selected	LLM: Observation note	**0.225**	**0.219**
Reference	Random assays + mean	0.141	0.303
Reference	Uncertainty + ridge	**0.152**	**0.292**
Reference	Random assays + ridge	0.141	0.303
Same assays	LLM: Hypothesis note	0.045	0.399
Same assays	LLM: Observation note	**0.064**	**0.380**
Shuffled feedback	LLM: Hypothesis note	0.018	0.426
Shuffled feedback	LLM: Observation note	**0.032**	**0.412**

H.3 Primary CRISPRa CD4 IL2

Comparison	Prediction procedure	n	MSE_0	$\text{MSE}_8 \downarrow$	D_8
LLM-selected	LLM: Hypothesis note	5	0.060	**0.061**	-0.001
LLM-selected	LLM: Observation note	5	0.074	0.062	0.002
Reference	Random assays + mean	5	0.057	0.063	0.000
Reference	Uncertainty + ridge	5	0.057	0.063	-0.000
Reference	Random assays + ridge	5	0.057	**0.061**	0.000
Same assays	LLM: Hypothesis note	5	0.061	**0.057**	0.003
Same assays	LLM: Observation note	5	0.079	0.063	-0.002
Shuffled feedback	LLM: Hypothesis note	5	0.063	**0.057**	0.003
Shuffled feedback	LLM: Observation note	5	0.059	0.059	0.001

Comparison	Prediction procedure	Final effect ↑	Regret ↓
LLM-selected	LLM: Hypothesis note	0.239	0.388
LLM-selected	LLM: Observation note	**0.357**	**0.270**
Reference	Random assays + mean	**0.339**	**0.288**
Reference	Uncertainty + ridge	0.227	0.400
Reference	Random assays + ridge	**0.339**	**0.288**
Same assays	LLM: Hypothesis note	0.204	0.423
Same assays	LLM: Observation note	**0.342**	**0.285**
Shuffled feedback	LLM: Hypothesis note	**0.077**	**0.550**
Shuffled feedback	LLM: Observation note	-0.016	0.643

H.4 Primary CRISPRa CD8 IFNG

Comparison	Prediction procedure	n	MSE_0	$\text{MSE}_8 \downarrow$	D_8
LLM-selected	LLM: Hypothesis note	5	0.070	0.063	0.013
LLM-selected	LLM: Observation note	5	0.061	**0.062**	0.014
Reference	Random assays + mean	5	0.067	**0.068**	0.000
Reference	Uncertainty + ridge	5	0.067	0.072	-0.001
Reference	Random assays + ridge	5	0.067	0.069	-0.001
Same assays	LLM: Hypothesis note	5	0.065	**0.063**	0.009
Same assays	LLM: Observation note	5	0.066	**0.063**	0.009
Shuffled feedback	LLM: Hypothesis note	5	0.067	**0.064**	0.007
Shuffled feedback	LLM: Observation note	5	0.063	0.066	0.005

Comparison	Prediction procedure	Final effect ↑	Regret ↓
LLM-selected	LLM: Hypothesis note	**0.882**	**0.165**
LLM-selected	LLM: Observation note	0.878	0.169
Reference	Random assays + mean	0.339	0.708
Reference	Uncertainty + ridge	**0.411**	**0.636**
Reference	Random assays + ridge	0.339	0.708
Same assays	LLM: Hypothesis note	**0.575**	**0.471**
Same assays	LLM: Observation note	0.542	0.504
Shuffled feedback	LLM: Hypothesis note	**0.577**	**0.470**
Shuffled feedback	LLM: Observation note	0.248	0.799

H.5 Primary CRISPRi CD4 IL2

Comparison	Prediction procedure	n	MSE_0	$\text{MSE}_8 \downarrow$	D_8
LLM-selected	LLM: Hypothesis note	5	0.049	0.021	-0.001
LLM-selected	LLM: Observation note	5	0.034	**0.020**	-0.000
Reference	Random assays + mean	5	0.017	0.018	0.000
Reference	Uncertainty + ridge	5	0.017	**0.017**	-0.000
Reference	Random assays + ridge	5	0.017	0.018	-0.000
Same assays	LLM: Hypothesis note	5	0.062	0.024	-0.002
Same assays	LLM: Observation note	5	0.035	**0.018**	0.002
Shuffled feedback	LLM: Hypothesis note	5	0.050	0.021	-0.001
Shuffled feedback	LLM: Observation note	5	0.060	**0.019**	-0.000

Comparison	Prediction procedure	Final effect ↑	Regret ↓
LLM-selected	LLM: Hypothesis note	0.273	0.363
LLM-selected	LLM: Observation note	**0.276**	**0.360**
Reference	Random assays + mean	0.220	0.416
Reference	Uncertainty + ridge	**0.313**	**0.323**
Reference	Random assays + ridge	0.220	0.416
Same assays	LLM: Hypothesis note	0.168	0.468
Same assays	LLM: Observation note	**0.228**	**0.408**
Shuffled feedback	LLM: Hypothesis note	**0.038**	**0.599**
Shuffled feedback	LLM: Observation note	0.019	0.617

H.6 Supplementary CRISPRa CD4 IFNG

Comparison	Prediction procedure	n	MSE_0	$\text{MSE}_8 \downarrow$	D_8
LLM-selected	LLM: Hypothesis note	5	0.045	0.041	0.002
LLM-selected	LLM: Observation note	5	0.050	**0.040**	0.002
Reference	Random assays + mean	5	0.039	0.041	0.000
Reference	Uncertainty + ridge	5	0.039	**0.040**	0.000
Reference	Random assays + ridge	5	0.039	**0.040**	0.000
Same assays	LLM: Hypothesis note	5	0.053	**0.042**	0.002
Same assays	LLM: Observation note	5	0.048	0.043	-0.000
Shuffled feedback	LLM: Hypothesis note	5	0.044	**0.041**	0.003
Shuffled feedback	LLM: Observation note	5	0.055	0.046	-0.003

Comparison	Prediction procedure	Final effect ↑	Regret ↓
LLM-selected	LLM: Hypothesis note	0.361	0.376
LLM-selected	LLM: Observation note	**0.369**	**0.368**
Reference	Random assays + mean	**0.304**	**0.433**
Reference	Uncertainty + ridge	0.223	0.514
Reference	Random assays + ridge	**0.304**	**0.433**
Same assays	LLM: Hypothesis note	**0.220**	**0.517**
Same assays	LLM: Observation note	0.101	0.636
Shuffled feedback	LLM: Hypothesis note	-0.059	0.796
Shuffled feedback	LLM: Observation note	**-0.039**	**0.777**

H.7 Supplementary CRISPRa CD4 IL2

Comparison	Prediction procedure	n	MSE_0	$\text{MSE}_8 \downarrow$	D_8
LLM-selected	LLM: Hypothesis note	5	0.124	**0.103**	0.005
LLM-selected	LLM: Observation note	5	0.119	0.104	-0.004
Reference	Random assays + mean	5	0.092	0.093	0.000
Reference	Uncertainty + ridge	5	0.092	0.094	0.000
Reference	Random assays + ridge	5	0.092	**0.092**	0.001
Same assays	LLM: Hypothesis note	5	0.131	0.110	-0.006
Same assays	LLM: Observation note	5	0.124	**0.109**	-0.007
Shuffled feedback	LLM: Hypothesis note	5	0.132	0.105	-0.002
Shuffled feedback	LLM: Observation note	5	0.124	**0.100**	-0.000

Comparison	Prediction procedure	Final effect ↑	Regret ↓
LLM-selected	LLM: Hypothesis note	0.340	0.383
LLM-selected	LLM: Observation note	**0.374**	**0.349**
Reference	Random assays + mean	**0.407**	**0.316**
Reference	Uncertainty + ridge	0.243	0.480
Reference	Random assays + ridge	**0.407**	**0.316**
Same assays	LLM: Hypothesis note	0.163	0.560
Same assays	LLM: Observation note	**0.446**	**0.277**
Shuffled feedback	LLM: Hypothesis note	-0.222	0.945
Shuffled feedback	LLM: Observation note	**-0.048**	**0.771**

H.8 Inference resources and recovery

Recorded response charges are $27.8130; conservative budget accounting totals $28.0603. The difference includes $0.2473 for 5 unanswered attempts after a disk-full interruption. Logs were preserved, validated, and resumed with exact-request caching; 10 episodes include cached calls and are excluded from uncached episode summaries. There were 49 malformed responses across 38 episodes. The shared repair policy exhausted both attempts on 5 steps, which used the prespecified deterministic fallback. All remain in the analysis. Primary probe requests required 0 fallbacks. Uncached LLM decision time has median 2.89s and p95 4.38s; full episode time has median 38.15s and p95 64.26s. CPU decision p95 is 0.687ms. The runner used an ARM Mac with macOS 15.6, six concurrent episodes, and remote Prime inference; provider hardware is undisclosed. No dedicated GPU was rented.

Prompt	Input tokens	Output tokens	Recorded study cost
Observation note	6,524,698	148,054	$13.481
Hypothesis note	6,556,069	152,740	$13.582

The feedback control uses separate temperature-zero API runs. Initial outputs can differ despite identical inputs, so final-error differences alone do not isolate feedback from serving variability. The accompanying summary reports initial differences and differences in learning gains as well as final errors.

Exploratory matching-mean reference. After inspecting the primary contrasts, we additionally evaluated the fixed average of the two disclosed single responses, corresponding to the matching-mean reference in Systema [Viñas Torné et al., 2025]. Its Norman MSE is 1.496, with split range [1.141, 1.737]. This post hoc reference does not tune its coefficient on held-out responses and is separate from the frozen baseline set.